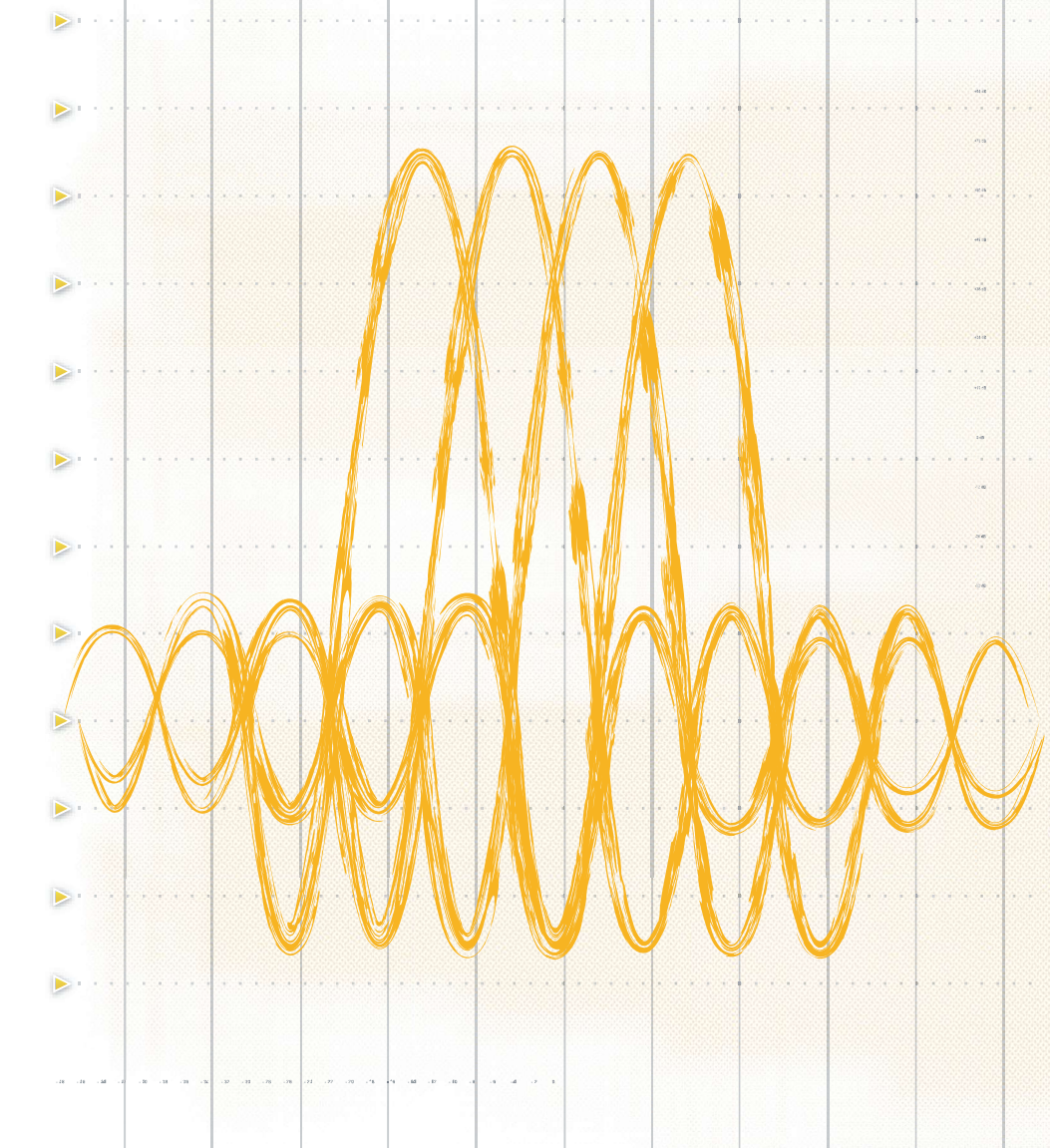




DIGITAL COMMUNICATIONS

PHYSICAL LAYER EXPLORATION LAB USING THE NI USRP™ PLATFORM

Robert W. Heath Jr., Ph.D., PE.

Digital Wireless Communication

Physical Layer Exploration Lab Using the NI USRP

Student Lab Manual

Robert W. Heath Jr.



ISBN-10: 1-934891-18-5
ISBN-13: 978-1-934891-18-6

10 9 8 7 6 5 4 3 2

Publisher: Tom Robbins
General Manager: Erik Luther
Technical Oversight: Sam Shearman
Marketing Manager: Amee Christian
Development Editor: Catherine Peacock
Compositor: Paul Mailhot, PreTeX Inc.

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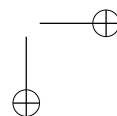
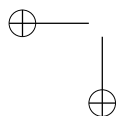
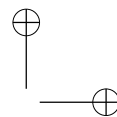
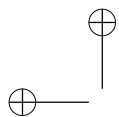
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Dedication

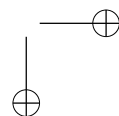
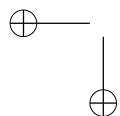
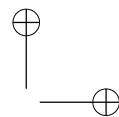
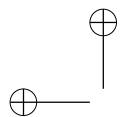
I dedicate this to Garima, Pia, and Rohan for their love and support,
to my parents Bob and Judy Heath for their encouragement
and to Dr. Mary Bosworth for her passion in the pursuit of higher education.

R. W. H.



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Preface

Wireless communication is fundamentally the art of communicating information without wires. In principle, wireless communication encompasses any number of techniques including underwater acoustic communication, semaphores, smoke signals, radio communication, and satellite communication, among others. The term was coined in the early days of radio, fell out of fashion for about fifty years, and was rediscovered during the cellular telephony revolution. Wireless now implies communication using electromagnetic waves—placing it squarely within the domain of electrical engineering.

Wireless communication techniques can be classified as either analog or digital. The first commercial systems were analog including AM radio, FM radio, television, and first generation cellular systems. Analog communication is rapidly being replaced with digital communication. The fundamental difference between the two is that in digital communication, the source is assumed to be digital. Modern applications of digital communication include cellular communication, wireless local area networking, personal area networking, and high-definition television.

Digital communication is widely taught in electrical engineering programs, at both the undergraduate and the graduate levels. Many digital communication courses take an abstract approach that emphasizes theory. This is no surprise—the mathematical theory of digital communication is beautiful and elegant. There has been a growing separation, however, in the abstract content of a typical course and what is implemented in practice. This difference is particularly evident in wireless communication systems, which have their own set of specific challenges. As courses are becoming more abstract, there is also increasing reliance on simulation to validate the theory. Typical simulations, though, are constructed under certain simplifying assumptions, which leaves important practical issues undiscovered. As a byproduct of the growing rift between theory and practice, and the increasing reliance on simulation, many students are not prepared to apply their knowledge immediately to a real digital wireless communication system.

This book is inspired by my own experiences as a graduate student and working at a startup company. My first summer project at Stanford University working with Arogyaswami Paulraj was to implement the physical layer of the receiver for a GSM system. The idea was to implement different space-time processing algorithms developed in Prof. Paulraj’s Smart Antennas Research Group and to show practically how multiple antennas can improve system performance using real transmitted waveforms. I found the implementation of the space-time processing algorithms to be straightforward (not easy but for the most part following the theory). Most of my time was spent, however, dealing with more practical issues like channel estimation, frame synchronization, and carrier frequency synchronization. I found that the algorithms for dealing with these practical impairments were vitally important yet I had never encountered them in multiple courses on digital communication and wireless communication systems.

I used what I had learned later at a startup company then called Gigabit Wireless and later Iospan Wireless, to develop a prototype of a three transmit and three receive antenna MIMO (multiple input multiple output) communication system. This prototype was key to demonstrating the promise of MIMO communication, which in 1998 was not yet an established hot research area. Coincidentally this prototype used multichannel DAC and ADC products from National Instruments.

As a new Assistant Professor at The University of Texas at Austin, I longed to teach wireless communication from a practical perspective. I quickly found one reason why these concepts were not taught in the typical communication course—it was not included in most textbooks. As a result, I developed course notes that include these concepts and will be published in a forthcoming textbook [3]; the key concepts are included in this lab manual.

I did not just want to teach wireless communication including some theory about dealing with practical impairments. I wanted to teach a class where students would send and receive actual communication waveforms. Unfortunately, when I started developing the course materials in 2003 there were few viable solutions for the radio frequency components to create a repeatable and reliable system. All of this changed with the introduction of hardware platforms with flexible RF upconverters and downconverters, matched to high speed DACs and ADCs that facilitate the concept of software defined radio. Finally there was a way to create and process waveforms with high quality and reproducibility. Using such equipment, the students could focus on the algorithms and yet be exposed to practical wireless system engineering, without having to worry with the challenging analog front end and RF circuit design issues. The present book makes use of National Instrument’s Universal Software Radio Peripheral (USRP) [13]. This low-cost hardware allows Universities to set up mul-

multiple workstations with a moderate budget. I am thrilled that this option has become available.

This book is a result of five years of development to make a complete laboratory course that teaches the principles of wireless digital communication. It consists of a series of labs with three components: a pre-lab, a laboratory experiment, and a lab report. The idea of the pre-lab is to write the essential LabVIEW code and getting it to work in a simulator before doing the laboratory experiment. This models conventional design practices where communication algorithms are tested in simulation before being tested over a wireless link. It is important to complete the pre-lab prior to starting the laboratory experiment. The laboratory experiments were designed to be completed in a time period of up to three hours. In the lab, the code from the pre-lab is run over the wireless link, some experiments are performed, and the results are recorded. The lab report is the final component of the lab. The purpose of the lab report is to discuss what was observed in the lab and to answer several questions related to wireless communication engineering. The lab report is an opportunity to synthesize what was learned.

Digital signal processing (DSP) is at the heart of the approach taken in this lab manual. No background in digital communication is assumed, though it would be helpful. The utility of a DSP approach is due to the bandlimited nature of wireless systems. Consequently with a high enough sampling rate, thanks to Nyquist’s theorem, it is possible to represent the bandlimited continuous-time wireless channel from its samples. This allows the transmitted signal to be represented as a discrete-time sequence, the channel as a discrete-time linear time-invariant system, and the received signal as a discrete-time sequence.

The labs explore both single carrier and multicarrier transmission. Single carrier transmission uses quadrature amplitude modulation (QAM) and raised-cosine pulse-shaping. Over the course of the labs, complexity is added to the receiver design including functions like detection, channel estimation, equalization, frame synchronization, and carrier synchronization. In later labs, the system is extended to incorporate multicarrier modulation in the form of orthogonal frequency division multiplexing (OFDM) with coding.

This lab manual could not have been possible with the support from many graduate teaching assistants at The University of Texas at Austin including Roopsha Samanta, Sachin Dasnurkar, Ketan Mandke, Hoojin Lee, Josh Harguess, Caleb Lo, Robert Grant, Ramya Bhagavatula, Omar El Ayache, Harish Ganapathy, Zheng Li, and Tom Novlan. Some specific contributions deserve special mention. Roopsha Samanta developed the initial set of laboratory drafts and software in addition to being the first TA for the course. Ketan Mandke reworked the software to have a complete system implemented and a channel simulator, where components could be removed and replaced with

software from the student. He also revised the lab manual to reflect the new system implementation. Hoojin Lee undertook the job of creating figures for the early course material. Josh Harguess created a nice introduction to LabVIEW presentation. Robert Grant spent time seeing how to convert to an early version of the USRP platform, when it was just released. Caleb Lo, Ramya Bhagavatula, Omar El Ayache, and Tom Novlan all made important revisions to the notes and software to keep it up-to-date with current versions of LabVIEW. All the teaching assistants ran the labs, worked with the students, collected feedback, and make suggestions and revisions. I sincerely thank them for their support.

The development of this book and course was supported by several groups. National Instruments made the course possible through its establishment of the Truchard Wireless Lab at The University of Texas at Austin. NI provided summer support that funded the endeavor, and contributed countless hours of employee time. Several groups from University of Texas at Austin including the Department of Electrical Engineering and the Cockrell School of Engineering provided teaching assistant support to develop and maintain the course materials. Working on the labs has inspired my research. I am pleased to acknowledge research support from the National Science Foundation, DARPA, the Office of Naval Research, the Army Research Labs, Huawei, National Instruments, Samsung, Semiconductor Research Corporation, Freescale, Andrew, Intel, and Cisco. Their support has allowed me to train many graduate students who share my appreciation of theory and practice.

I hope you enjoy this book and doing the laboratory experiments.

ROBERT W. HEATH JR.
Austin, Texas

About the Author

Robert W. Heath Jr. is an Associate Professor in the Department of Electrical and Computer Engineering at The University of Texas at Austin and a member of the Wireless Networking and Communications Group. He is also President and CEO of MIMO Wireless Inc. and VP of Innovation at Kuma Signals LLC. He received his B.S. and M.S. degrees from the University of Virginia, Charlottesville, VA, in 1996 and 1997 respectively, and the Ph.D. from Stanford University, Stanford, CA, in 2002, all in electrical engineering. Prior to joining The University of Texas, he was with Iospan Wireless Inc, San Jose, CA where he worked on the design and implementation of the physical and link layers of the first commercial MIMO-OFDM communication system.

His current research interests include several aspects of wireless communication and signal processing including limited feedback techniques, multihop networking, multiuser and multicell MIMO, interference alignment, adaptive video transmission, manifold signal processing, and 60 GHz communication techniques. He has published some 300 papers in archival journals and conference proceedings on these topics, and has more than 25 U.S. patents.

He has been an Editor for the IEEE Transactions on Communication, an Associate Editor for the IEEE Transactions on Vehicular Technology, lead guest editor for an IEEE Journal on Selected Areas in Communications special issue on limited feedback communication, and lead guest editor for an IEEE Journal on Selected Topics in Signal Processing special issue on Heterogenous Networks. He currently serves on the steering committee for the IEEE Transactions on Wireless Communications. He was a member of the Signal Processing for Communications Technical Committee in the IEEE Signal Processing Society. Currently he is the Chair of the IEEE COMSOC Communications Technical Theory Committee. He was a technical co-chair for the 2007 Fall Vehicular Technology Conference, general chair of the 2008 Communication Theory Workshop, general co-chair, technical co-chair and co-organizer of the 2009 IEEE Signal Processing for Wireless Communications Workshop, local co-organizer for the 2009 IEEE CAMSAP

Conference, technical co-chair for the 2010 IEEE International Symposium on Information Theory, technical chair for the 2011 Asilomar Conference on Signals, Systems, and Computers, and is technical co-chair for the 2014 IEEE GLOBECOM conference.

He was a co-author of best student paper awards at IEEE VTC 2006 Spring, WPMC 2006, IEEE GLOBECOM 2006, IEEE VTC 2007 Spring, and IEEE RWS 2009. He is co-recipient of the Grand Prize in the 2008 WinTech WinCool Demo Contest and is co-recipient of the 2010 EURASIP Journal on Wireless Communications and Networking best paper award. He was a 2003 Frontiers in Education New Faculty Fellow. He is the recipient of the David and Doris Lybarger Endowed Faculty Fellowship in Engineering. He is a licensed Amateur Radio Operator and is a registered Professional Engineer in Texas. He is a Fellow of the IEEE.

Lab 1: Introduction to Wireless Communications Lab Part 1: Introduction to NI LabVIEW

Summary

A software defined radio (SDR) is a communication radio that uses software to implement the algorithms necessary for digital communication. In this lab, you will be designing and implementing an SDR using National Instruments hardware and software (i.e., National Instruments LabVIEW). Through the implementation of this SDR you will investigate practical design issues in wireless digital communication.

This introductory lab is divided into two parts. In the first part you will learn about the software used in this lab; in the second part you will learn about the RF hardware to be used in lab. This part of the lab will answer the following questions:

- What software will I need for this lab?
- What is NI LabVIEW and why do we use it in this lab?
- What are some of the advanced LabVIEW tools used in this lab?
- Is there such a thing as good code style in LabVIEW?

Also, in this part of the lab, you will build some of the basic components of the SDR that you will be designing throughout the labs. This lab is meant to introduce LabVIEW as a tool for the design of digital communications systems in this course, and to give you a chance to get comfortable with basic LabVIEW programming concepts.

1 Software and Other Materials for the Course

Throughout the course you will be required to do some portion of the lab work (e.g., pre-labs) outside of the lab. This means that you will need to get access to the appropriate software tools and packages outside of class. The NI software below will be used in the course.

- NI LabVIEW 2010 or later
- NI LabVIEW Modulation Toolkit 4.3.1 or later

Note: Your instructor will provide information about how to access LabVIEW at your University.

2 Introduction to National Instruments LabVIEW

2.1 What Is LabVIEW?

LabVIEW is a graphical programming language developed by National Instruments. The basic building block of LabVIEW is the virtual instrument (VI). Conceptually, a VI is analogous to a procedure or function in conventional programming languages. Each VI consists of a *block diagram* and a *front panel*. The block diagram describes the functionality of the VI, while the front panel is a top level interface to the VI. The construct of the VI provides two important virtues of LabVIEW: code reuse and modularity. The graphical nature of LabVIEW provides another virtue: it allows developers to easily visualize the flow of data in their designs. NI calls this Graphical System Design. Also, since LabVIEW is a mature data flow programming language, it has a wealth of existing documentation, toolkits, and examples which can be leveraged in development.

In this course you will use National Instruments RF hardware. LabVIEW provides a simple interface for configuring and operating various external I/O, including the NI RF hardware used in lab. This is the main reason why you will use LabVIEW as the programming language to build an SDR in this course. You should realize that the algorithms considered here could also be programmed in optimized C/C++, assembly, or VHDL and implemented on a DSP, microcontroller, or an FPGA. The choice of hardware and software in this lab is mostly a matter of convenience.

In future labs you will need to be familiar with LabVIEW and the documentation/help available to you. This is the only lab in this course which will give you the opportunity to learn and practice LabVIEW programming; so it is important that you take this opportunity to ask the instructor any questions you might have about LabVIEW programming. The following tutorials and reference material will help guide you through the process of learning LabVIEW:

- LabVIEW 101 [11],
- *LabVIEW Fundamentals* from National Instruments [12],

- and online LabVIEW tutorials from NI [9, 10].

New LabVIEW programmers should carefully review all of the material in [11] and [9]. Please remember to refer to [12] and [10] often as they are excellent references for all basic LabVIEW questions. In order to test your familiarity with LabVIEW, you will be asked at the end of this lab to build some simple VIs in LabVIEW and integrate them into a simulator. Bring any questions or concerns regarding LabVIEW or these tutorials to your instructor’s attention. For the remainder of this lab you should be familiar with the basics of LabVIEW programming and where to look for help.

2.2 What Advanced LabVIEW Tools Will I Use?

Many of the algorithms you will implement in the lab (and in digital communication in general) use linear algebra. LabVIEW provides a variety of support for matrix and vector manipulation and linear algebra. Some of the functions you will use regularly include matrix inversion (*Inverse Matrix.vi*), matrix multiplication (*A x B.vi*), and reshaping arrays (*Reshape Array.vi*). LabVIEW also has many built-in signal processing functions which will be used in the lab, such as the fast Fourier transform (*FFT.vi*), inverse FFT (*Inverse FFT.vi*), and convolution (*Convolution.vi*). The Modulation Toolkit is a toolset of common digital communication algorithms which will also be leveraged in this lab. Note that many of the functions you will implement in this lab are already available in the Modulation Toolkit in some form. The objective of this course is to understand the principles of wireless digital communication by implementing the physical layer in as much detail as possible. Once you are familiar with these concepts, you will decide when to use existing VIs and when to write your own. It is recommended that you explore the following tool palettes in order to get some exposure to the VIs that you will likely use in this course:

- Signal Processing palette
- Digital palette (part of the Modulation Toolkit palette)
- Structures palette
- Complex palette (part of the Numeric palette)
- and Array palette

Appendix A enumerates some common VIs to be used in this course and how to access them.

2.3 Code Style in LabVIEW

Googling the phrase “*labview code style*” will yield a number of sources for general code style in LabVIEW. Whatever style you choose to follow, remember that your code style should accomplish the following:

1. Consistency - All of your code should follow the same code style.
2. Readability - Your code should be user-friendly and easy to maintain.
3. Documented - Documentation should not be few and far between. Good documentation not only comes in the form of free text labels (see the Decorations palette, which is part of the Structures palette), but it also means good variable and function names.

When incrementally designing a moderately complicated system, it can be tempting to continually add *hacks* until you end up with *spaghetti code* (see Figure 1).

The design of a digital communication system is modular and sequential in nature. Thus it is natural to adopt a code style which makes use of a great deal of hierarchy and modularity. Since many algorithms in digital communications can often be boiled down to a recipe (*e.g.*, *A then B then C*), it is critical to use sub-VIs whenever possible.

3 Pre-Lab

In this pre-lab you will build two sub-VIs: *source.vi* and *error_detect.vi*. You will then plug your code into a simulator provided to you (shown in Figure 2). The goals of this pre-lab are (1) to understand how to build and use sub-VIs and (2) to practice using good code style.

The details of each sub-VI that you must construct are given in Tables 1 and 2. In all future labs you will be provided with an appropriate simulator. You will typically reimplement certain blocks within the simulator and use it to perform some kind of measurement. This incremental process will continue throughout the course as you learn about each part of a digital communication system. The general simulator you will use in future labs is composed of many files. Figure 3 shows an overview of the hierarchy of the various parts of the simulator.

It should be noted that the simulator you will be provided with will have full functionality; however, the block diagram of the functional blocks of the simulator will be locked. This means that although you will be able to see how the simulator is supposed to function and look at the front panel of all the VIs, you will not be able to see the block diagram (*i.e.*, the implementation) of the digital communication sub-VIs.

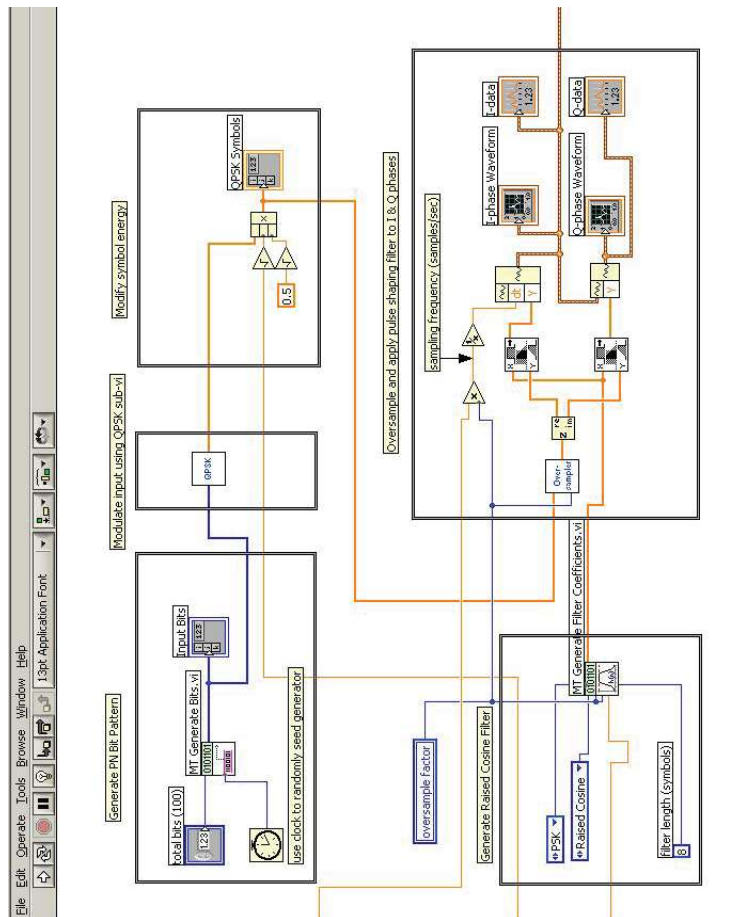


Figure 1: Spaghetti code design can be messy.

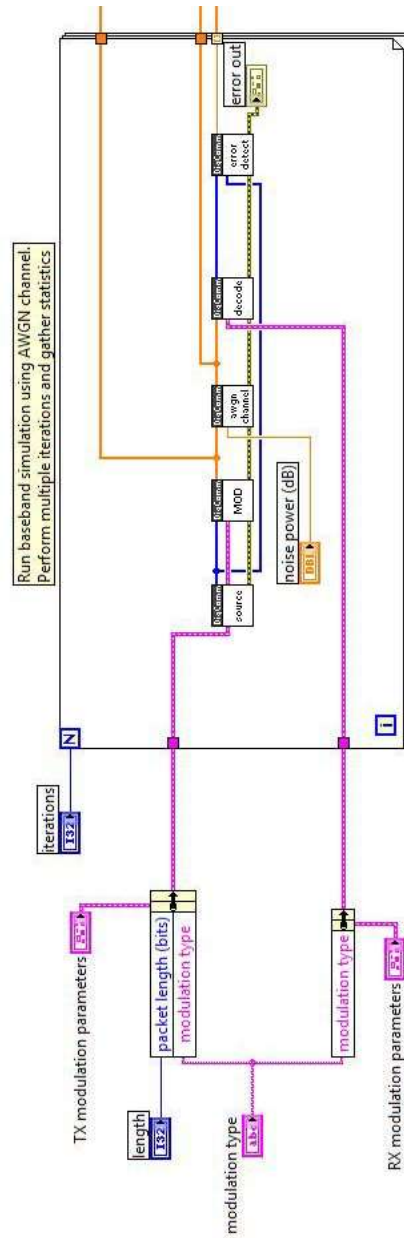


Figure 2: Digital communications simulator. Block diagram of *awgn_simple_sim.vi*.

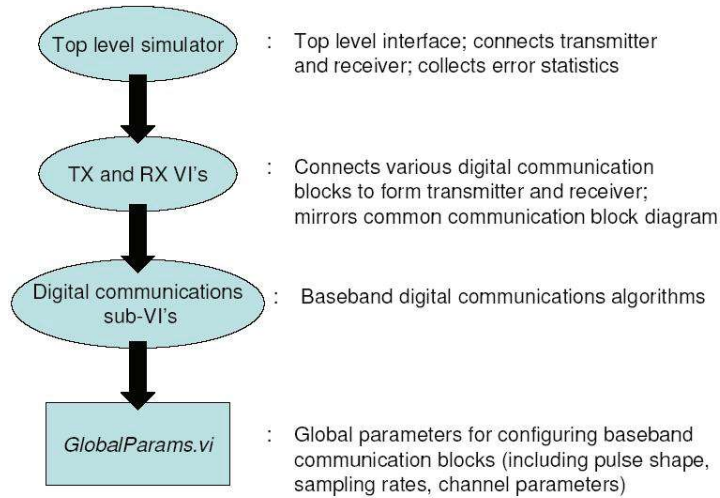


Figure 3: Overview of simulator hierarchy in LabVIEW.

Table 1: Description of *source.vi*

<i>source.vi</i> - generates a random bit sequence ¹			
INPUTS	<i>length</i>	I32 (32 bit integer)	length of bit sequence
OUTPUTS	<i>output bit sequence</i>	array of U8 (unsigned bytes)	random bit sequence (array of unsigned bytes taking value 0 or 1)

Once you have constructed the VIs for this lab (*source.vi* and *error_detect.vi*), be sure to appropriately edit the icon of each VI and connect all of the relevant terminals. Now you are ready to replace the corresponding blocks inside the simulator provided to you (shown in Figure 2). After placing your code in the simulator and verifying that it works correctly, observe what happens to the bit-error rate as you decrease the noise power in the simulator. (Note: If your subVI appears greyed out after placing it in the simulator, right click the subVI and choose "relink subVI" to fix this issue)

¹Hint: There are a number of ways to generate a random bit sequence in LabVIEW. For instance, you could do it using the random number generator provided in the Numeric palette, or you could just use the Generate Bits.vi module located in the Digital palette (see Modulation Toolkit palette).

Table 2: Description of *error_detect.vi*

<i>error_detect.vi</i> – computes the bit error rate of a bit sequence			
INPUTS	<i>estimated bit sequence</i>	array of U8	estimated bit sequence
	<i>key bit sequence</i>	array of U8	correct bit sequence
OUTPUTS	<i>bit-error rate</i>	DBL (double precision)	Hamming distance ² between estimated sequence and key sequence divided by length of bit sequence

The simulator will allow you to run multiple iterations and collect statistics from these runs. In this case, you will only be concerned with the bit-error rate (BER) statistics (i.e., the average BER).

Plot a curve of the average bit-error rate versus $1/N_0$ over a number of trials, where N_0 is the noise power (realize N_0 is given in dB in the simulator). It is sufficient to vary noise power from -10 dB to 0 dB in 2 dB steps. Use BPSK modulation and choose an input length of 200 bits. *Note that in order to observe a nonzero error for low noise powers you may need to run in excess of 10^4 iterations.*

HINT: (X/Y) in dB is equal to $10 \log \left(\frac{X_{\text{linear}}}{Y_{\text{linear}}} \right)$, where $X_{dB} = 10 \log(X_{\text{linear}})$ and $Y_{dB} = 10 \log(Y_{\text{linear}})$. The $(\cdot)_{\text{linear}}$ is a linear scale value, not dB.

Pre-Lab Turn In

Submit your implementations of *source.vi* and *error_detect.vi*. Turn in your plots of average bit-error rate (BER) versus $1/N_0$.

A typical example of a bit-error-rate plot is shown in Figure 4. You may use LabVIEW, Excel, or any other graphing utility to generate these plots (i.e., you only need to use LabVIEW to generate the data, not necessarily to generate the plots).

²The Hamming distance of two bit sequences is equal to the number of positions in which the sequences differ.

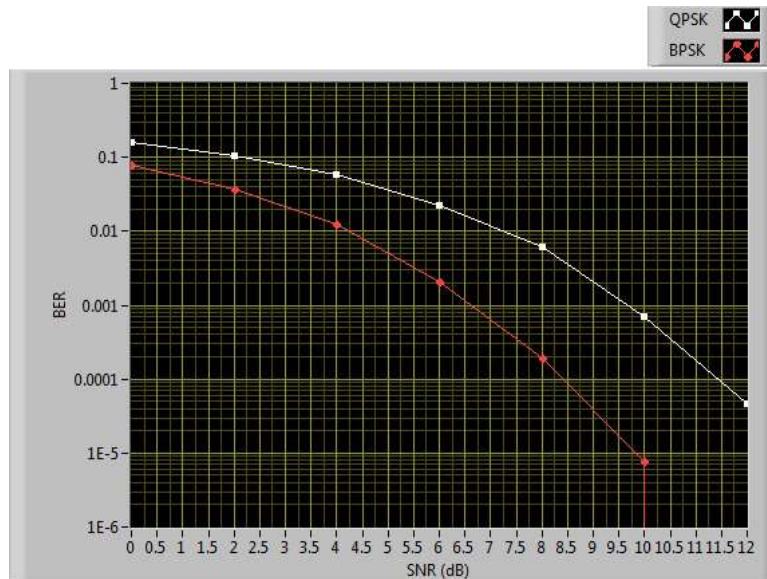


Figure 4: BER plot for BPSK and QPSK modulation in AWGN channel.

4 Lab Experiment

There is no in-lab component for this lab.

5 Lab Report

After most labs you will write a lab report. You will only turn in one lab report per lab. This means that if a lab has multiple parts (e.g., Parts 1 and 2 of Lab 1), you will only be required to turn in one lab report. The contents of this report will be explicitly stated in the lab draft. *Note: This means that you will not be submitting a lab report for this part of Lab 1.* Upon completion of Part 2 of Lab 1, you will be required to generate a lab report.

Lab 1: Introduction to Wireless Communications Lab Part 2: Introduction to NI RF Hardware

Summary

In this part of the introductory lab you will learn about the NI RF hardware to be used in the Wireless Communications Lab. You will acquaint yourself with the relevant documentation and references available to you with regard to the RF hardware. You will build some simple example VI's from the NI documentation and use some of the existing LabVIEW tools (e.g., RF Signal Analyzer Demo Panel). In the final part of this lab you will explore some of the VIs that you will use in future labs to control the RF hardware. This part of Lab 1 introduces NI RF hardware as a tool for digital communication, and gets you familiar with the help/documentation available for the hardware you will be working with throughout the course.

In this lab, for the pre-lab, you are required to turn in only the questions given in Section 2. There are no pre-lab VIs for this lab. As a part of the lab session, you will construct two VIs by following the instructions given in the NI documentation. The questions given in Section 3 need to be answered during the lab only. Please do not submit the answers to these questions with your pre-lab. At the end of this lab, you will submit a lab report, the details of which are given at the end of this document.

1 RF Hardware in Wireless Communications Lab

In this course you will use software reconfigurable RF hardware from National Instruments to build a digital communication system. This hardware can be easily configured using LabVIEW. The RF Hardware used in the lab is the National Instruments USRP (Universal Software Radio Peripheral). Figure 1 shows a USRP connected to a PC (running LabVIEW). This PC controls the USRP through the gigabit ethernet cable connecting the two together. In this section you will learn more about these modules.

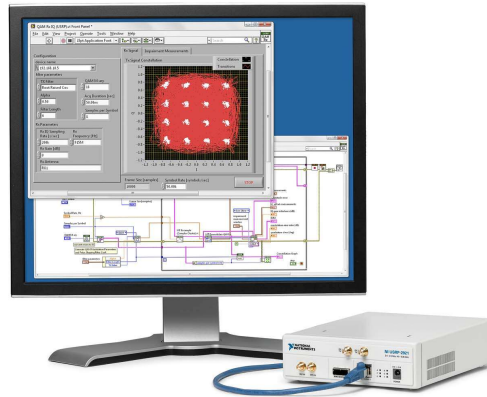


Figure 1: Hardware setup in Wireless Communications Lab.

1.1 NI USRP (Transmit)

Let's begin by taking a look at the NI USRP when acting as a transmitter. The USRP receives a waveform from the host PC with 16 bits of resolution sampled at up to 25 MSamples/second. This signal is upconverted to a radio frequency (RF) before being sent to an amplifier and then transmitted over the air. For more information on the NI USRP, please refer to the help files associated with the USRP (**Help⇒NI-USRP Help⇒Devices⇒NI USRP-292x Specifications**)

1.2 NI USRP (Receive)

The NI USRP is also capable of receiving, and does so in the following manner. The received signal is mixed with a desired carrier frequency in order to down-convert it to a complex IQ baseband signal sampled at 100 MSamples/second. The digital signal is then downsampled to a rate specified by the user and passed to the host PC for processing.

When the ADC of the NI USRP samples at the full digitizer rate (100 MSamp/sec), it can acquire a bandwidth of 20 MHz. Sampling at such a high rate with high resolution (14 bits) produces a large amount of data. However, to acquire a signal with smaller bandwidth (a narrowband signal), it is sufficient to sample at a rate twice that of the signal bandwidth¹.

¹This sampling rate is known as the Nyquist rate of the system. It is the minimum sampling rate required to reproduce a bandlimited signal from discrete samples.

Decimation is the process of reducing the sampling rate of a discrete-time signal. In practice, this entails applying an anti-aliasing filter and throwing away some samples. The NI USRP features a digital down converter (DDC), which allows it to perform decimation in hardware instead of in software (which can be considerably slower).

Using the DDC, the USRP can acquire narrowband signals at sample rates much less than the full digitizer rate (100 MSamp/sec), thus reducing the memory required to store a waveform. For example, a waveform sampled at 100 MSamp/sec for 1 second produces 100 times more data than the same waveform sampled at 1 MSamp/sec. The effective sample rate of the USRP (including DDC) ranges from a maximum of 25 MSamp/sec down to 200 kSamp/sec and is directly configurable by the user. For more information on the NI USRP, please refer to the associated help files.

2 Pre-Lab

Review the documentation for the USRP (**Help⇒NI-USRP Help**). You will find that the help files found under **Devices⇒NI USRP-292x Specifications** and **Devices⇒NI USRP-2920** contain the most comprehensive information about the USRP. For a deeper understanding of the specifications, please refer to the following documents:

- Bandwidth, Sample Rate and Nyquist Theorem [8]
- Understanding RF & Microwave Specifications - Part I [14]
- Understanding RF & Microwave Specifications - Part II [15]

Answer the following questions about the USRP after reading through the help files.

1. What is the range of allowable carrier/center frequency supported by the NI-USRP?
2. What is the maximum allowable bandwidth supported by the NI-USRP?
3. What is the maximum sampling rate of the NI-USRP?
4. Why do you think the DDC is implemented? What is its main benefit?

Answer the following general concept questions using [8], [14] and [15].

1. In your own words, describe what the bandwidth of an instrument is.

2. What is meant by the sampling rate of an instrument?
3. Why are these specifications important for designing a transmitter and receiver in a wireless communications system?

Pre-Lab Turn In

Submit your answers for all of the above pre-lab questions. As a reminder, this is all that you need to turn in for the pre-lab. There are no pre-lab VIs to be submitted for this lab.

3 Lab Experiment

3.1 Building a Basic Sine Generation VI

For the first task in this lab you will build a basic sine generation VI described below. After you have completed building this example VI, verify that the block diagram of your VI resembles Figure 2, and that the front panel resembles Figure 3.

Use the following procedure to begin using the NI USRP to generate a signal:

1. Select **Start⇒Programs⇒National Instruments⇒LabVIEW 2010⇒LabVIEW** to launch LabVIEW. The LabVIEW dialog appears.
2. Click **New**, choose **Blank VI**, and click **OK** to create a blank VI.
3. Display the block diagram by clicking it or selecting **Window⇒Show Block Diagram**.
4. Navigate to the NI USRP VIs on the **Functions⇒Instrument I/O⇒Instrument Drivers⇒NI-USRP⇒TX** palette. With LabVIEW running, click here to find the **NI USRP TX** LabVIEW palette.
5. Create the block diagram shown in Figure 2 by placing the four core VIs on the block diagram in the order they appear in the NI USRP RX functions palette.
6. Hover the cursor over the **device name** terminal on the niUSRP Open TX Session VI and right-click. Select **Create⇒Control** to create a front panel field where you specify the NI USRP device name.
7. Hover the mouse tool over the **IQ rate**, **carrier frequency**, and **gain** terminals of the niUSRP Configure Signal VI.

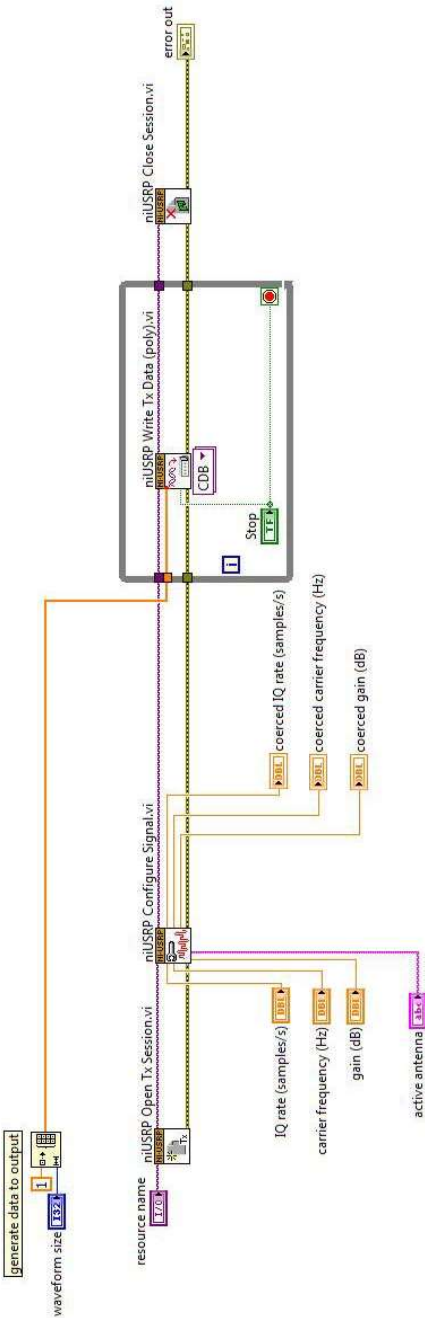


Figure 2: Block diagram of Basic Sine Wave Generation VI.

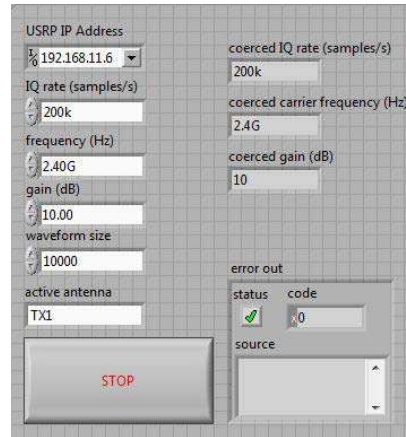


Figure 3: Front panel of *Basic Sine Wave Generation VI*.

8. Right-click each terminal and select **Create⇒Control** from the shortcut menu to create frequency, IQ rate, and gain controls.
9. Display the front panel by clicking it or selecting **Window⇒Show Front Panel**. Fields are displayed in which you can specify a frequency, IQ rate, and gain.

Continuous waveform generation is controlled by means of a STOP button. A STOP button is typically used within a While Loop. This example places a While Loop around the Write TX Data VI so that signal generation continues until you click STOP.

Build a While Loop and STOP button by completing the following steps:

1. Display the block diagram by clicking it or selecting **Window⇒Show Block Diagram**.
2. Select the While Loop on the **All Functions⇒Structures** palette. With LabVIEW running, click here to find the While Loop on the **Structures** palette.
3. Enclose the niUSRP Write TX Data (Poly) VI in the While Loop.
4. Hover the mouse tool over the **Loop Condition** terminal of the While Loop.
5. Right-click the **Loop Condition** terminal and select **Create Control** from the shortcut menu to create a **STOP** button on the VI front panel.

6. Create an error indicator by right-clicking on the **error out** terminal of the niUSRP Close Session VI and selecting **Create⇒Indicator**.

Now we need to produce a waveform to be transmitted. We will be transmitting a sinusoid at a carrier frequency that is specified by the **carrier frequency** control. This means that the baseband signal we will be passing to the NI USRP is simply a constant signal. To generate this signal, complete the following steps:

1. Using the quick-drop functionality described in Lab 1, Part 1, place the **Initialize Array** function down on the block diagram.
2. Right-click the **element** input of the **Initialize Array** function and select **Create⇒Constant**. Double-click on the constant that was created and change it to 1.
3. Right-click on the **dimension size** input and select **Create⇒Control**.
4. Wire the output of the **Initialize Array** function block to the **data** input of the niUSRP Write Tx Data (poly) VI.
5. Finally, wire the STOP button that was created for the While Loop to the **end of data?** input of the niUSRP Write Tx Data (poly) VI and select **CDB** from that VIs drop down menu.

Remember that in order to operate your sine generation VI, you must properly configure the *USRP IP Address* for the USRP (an input to *niUSRP Initialize.vi*). You can find the IP addresses for all of the USRP devices connected to your PC by using the **NI-USRP Configuration Utility**. Once that is properly configured, set the rest of the parameters as follows and run the VI:

- carrier frequency = 2.4 GHz
- IQ rate (samples/s) = 400k
- gain (dB) = 20
- waveform size = 10000
- active antenna = TX1

You will now test your VI using the niUSRP EX Spectral Monitoring (Interactive) VI, which can be found in the Functions palette. This spectrum analyzer tool will allow you to observe the frequency domain response of your basic sine wave transmitted over a wireless link. The following directions describe the basic operation of the Spectral Monitoring VI.

1. First, open the Spectral Monitoring VI. This can be accessed from **Functions Palette**⇒**Instrument I/O**⇒**Instrument Drivers**⇒**NI-USRP**⇒**Examples**⇒**niUSRP EX Spectral Monitoring (Interactive).vi**.
2. Verify that the USRP IP Address match the values specified in the NI-USRP Configuration Utility.
3. After opening the example VI, you will want to modify the hardware settings on the front panel. Use the parameters listed below for this experiment.
 - Center Frequency = 2.4 GHz
 - IQ Sampling Rate [s/sec] = 400k
 - Acq Duration [sec] = 50m

Questions

Answer the following questions regarding the NI USRP.

1. What is the IP Address for your NI USRP?
2. Describe what happens to the spectral response of the transmitted signal when you place an obstruction between the antenna of transmitter and that of the receiver (i.e., blocking the line of sight component of the waveform).

Turn In

Demonstrate to the instructor that your basic sine generation VI is working properly. You will need to start the Spectral Monitoring VI and show that your basic sine generation VI produces a carrier signal at 2.4 GHz.

Submit your answers to the above questions in your lab report. More details about the requirements for the lab report are discussed near the end of this document.

3.2 Programming the RF Signal Analyzer

In the second part of this lab experiment you will build a Continuous IQ Acquisition VI and it should look like the Continuous IQ Acquisition VI shown in Figure 4. Add all of the appropriate controls and indicators necessary to control the sub-VIs in your code.

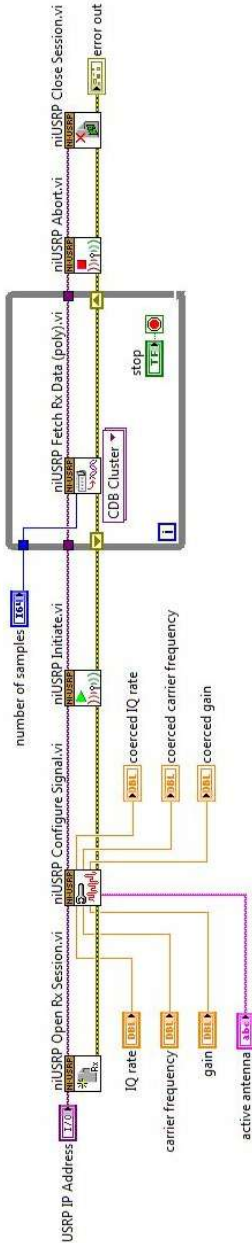


Figure 4: Continuous IQ Acquisition VI.

To test your acquisition VI, you will graph the real and complex parts of the signal acquired by *niUSRP Fetch Rx Data (poly).vi*. To do this:

1. Place two waveform graphs on the front panel of your VI. These can be accessed from the **All Controls** tool palette (**All Controls**⇒**Graph**⇒**Waveform Graph.vi**).
2. Retrieve the real and imaginary waveform components from the *IQ Waveform* output of *niUSRP Fetch Rx Data (poly).vi*. Use *Get Complex IQ Component.vi* to retrieve these components from *IQ Waveform*. This VI can be accessed from the **All Functions** tool palette on the block diagram (**All Functions**⇒**Modulation**⇒**Analog**⇒**Utilities**⇒**Get Complex IQ Component.vi**).
3. On the block diagram, connect the terminal for each waveform graph to its corresponding waveform for the real and imaginary parts of the acquired signal (inside of the While-Loop structure).

You will use the basic sine generation VI created in the previous section to test your acquisition VI. In order to do so, you must first set up the parameters for the receiving USRP. Set the input parameters for *niUSRP Configure Signal.vi* listed below. Notice the carrier signal generated by your basic sine generation VI will be offset from the center frequency of your acquisition VI, effectively creating a 100 kHz single-sideband (SSB) modulated waveform at baseband.

- Carrier Frequency = {2.4001 GHz @ transmitter} and {2.4 GHz @ receiver}
- IQ rate = 800k
- gain = 20 dB
- number of samples = 10000
- active antenna = RX1

You should now be ready to continuously acquire a signal. After setting the parameters described above, run both VIs and observe the in-phase and quadrature-phase waveforms (i.e., real and imaginary parts respectively) of the acquired signal.

Questions

Answer the following question about the NI USRP.

1. What are the three elements in the *data* cluster output from *niUSRP Fetch Rx Data (poly).vi*? Give the name and data type of each element.

Turn In

Demonstrate to the instructor that your acquisition VI is working properly. It must correctly display both I and Q phases of the acquired signal.

Submit your answers to the questions in your lab report.

3.3 Looking Ahead

For the rest of this course, VIs for RF hardware configuration and operation will be provided to you. This will allow you to focus on the details of the digital communications algorithms you will be implementing. Even though you will be provided with the VIs necessary for controlling the RF hardware, you will be expected to understand what these VIs do and how they work.

You will be using the VIs listed below in lab to configure and operate the RF hardware. Explore the front panels and block diagrams of these VIs.

- *TXRF_init.vi* - initializes a USRP transmit session with the appropriate RF parameters.
- *TXRF_send.vi* - executes a transmit operation (i.e., sends a complex waveform), and closes the USRP session.
- *RXRF_init.vi* - initializes an USRP receive session with the appropriate RF parameters.
- *RXRF_recv.vi* - acquires a complex waveform, and closes the USRP session.

Questions

Answer the following questions about the VIs you will be using in lab. *Hint: The help files for VIs can be accessed from the block diagram. Right-click on the VI you wish to find out more about and select “Help.”*

1. In *RXRF.config.vi* the **IQ rate** parameter is used by *niUSRP Configure Signal.vi*. Can the USRP support any arbitrary IQ rate? If not, what will the USRP do when an unsupported bandwidth parameter is requested?

Turn In

Submit all of your answers in your lab report.

DIRECTIONS FOR SUBMISSION OF THE LAB REPORT

Your lab report should accomplish the following tasks.

1. Answer all of the questions asked in lab (i.e., do not reanswer pre-lab questions).
2. Discuss problems you encountered in Lab 1, Part 1 and how you overcame these obstacles.
3. In 2-3 sentences describe the basic operation of the NI USRP when transmitting (e.g., what are the inputs/outputs for the USRP?).
4. In 2-3 sentences describe the basic operation of the NI USRP when receiving (e.g., what are the inputs/outputs for the USRP?).
5. Discuss problems you encountered in this part of Lab 1 and how you overcame these obstacles.